Forest-method and proximity-application roadmap

Decision ledger. Evidence and repository state frozen on **2026-07-14**. This is the implementation-facing companion to the literature ledger, not a claim that its 50 direct proximity works are an exhaustive final corpus. The ledger is a strong, checked snapshot; its counts, bibliography, and local-PDF inventory still need a reconciliation pass.

The unit of comparison is a scientifically valid cell:

```
estimator family × relation definition × application × data regime
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An estimator family is not a relation definition. For example, an RF can produce original co-leaf affinity, an OOB approximation, RF-GAP prediction weights, or a future KeRF normalisation; an isolation forest's native path-length score is not thereby an RF-GAP proximity.

Reading this ledger

Decision labels: reuse = supported here and stabilize; **integrate** = use an existing dependency behind an adapter; **modernize/port** = rehabilitate an existing implementation with an oracle; **build** = implement a missing relation/application; **specialty** = bounded, data-specific comparison; **reference-only** = document but do not make a required implementation.

“Available” below means available to this checkout, not merely installable. `treeple==0.10.3` is an optional research dependency in the `treeple` extra; it is noncommercial and must not be vendored. An unavailable optional dependency must remain an explicit unsupported registry row, rather than silently shrinking a study.

The 1.0 public composition surfaces are `Proximity`, `get_proximities`, and `prox_extend`. Current general relations are `original`, `oob`, `rfgap`, `honest_gap`, `kerf`, `path`, and `random_height`; method-specific relations are named `collision`, `conditional`, and `survival_leaf`. `honest_gap` is restricted to an honest forest. `rfgap` is asymmetric prediction-weight geometry, while the original relation is a symmetric co-leaf affinity. See the current application semantics and the manifold conversion tests.

Method decision matrix

Task support is C=classification, R=regression, U=unsupervised/anomaly, S=survival, and CA=causal. “Leaf/OOB contract” names the information an adapter must expose: leaf assignment (L), tree inventory (T), in-bag multiplicities (IB), OOB/query masks (O), and per-tree contribution weights (W).

Foundational family	Task / partition geometry	Python/sklearn status; other-language oracle	Leaf/OOB contract; license & maintenance	Scientifically valid relation(s)	Decision and rationale
CART	C/R; axis-aligned recursive regions	sklearn trees; R `rpart`	L/T only; BSD-3-Clause sklearn, maintained	Original co-leaf / leaf incidence only	**reference-only** ancestor; sklearn RF supplies the relevant ensemble baseline.
ID3	C; categorical information-gain regions	No project adapter; Weka ID3	L/T only; Weka GPL, maintained suite	Original co-leaf only	**reference-only** historical contrast.
C4.5	C; pruned axis-aligned regions with missing/categorical routing	No project adapter; Weka J48/C5.0	L/T only; external implementation; maintained descendants	Original co-leaf only	**reference-only**; unique missing-value routing is not currently a hypothesis.
Random subspace	C/R; feature-subsampled trees	Covered as RF component; Weka oracle	L/T; implementation dependent	Original leaf incidence	**reference-only** precursor; redundant with RF ablations.
Bagging	C/R; bootstrap-aggregated partitions	sklearn Bagging; no generic project adapter	L/T/IB/O required; BSD-3-Clause sklearn, maintained	Original/OOB only if bootstrap roles are recoverable	**reference-only** ancestor; needed conceptually for OOB/RF-GAP validity.
Random Forest	C/R; axis-aligned bootstrap forest	sklearn RF available	L/T/IB/O/W; BSD-3-Clause, maintained	Original, OOB, RF-GAP; future KeRF/path/random-height	**reuse** and stabilize; baseline implementation and prediction-identity anchor.
Extra-Trees	C/R; randomized axis-aligned cuts	sklearn ExtraTrees available	L/T; IB/O only under an explicitly valid bootstrap configuration; BSD-3-Clause, maintained	Original; OOB/RF-GAP only when bootstrap contract holds	**reuse** as randomized-partition comparator.

Foundational family	Task / partition geometry	Python/sklearn status; other-language oracle	Leaf/OOB contract; license & maintenance	Scientifically valid relation(s)	Decision and rationale
AdaBoost	C/R; sequential weighted trees	sklearn available	L/T/W; no exchangeable bootstrap/OOB contract	Explicit boosted-tree relation only	**reference-only**; ordinary RF-GAP is invalid.
Gradient boosting	C/R; additive stagewise trees	sklearn available	L/T/W; no RF bootstrap/OOB	Explicit contribution-weighted relation only	**reference-only** ancestor; not an RF-GAP target.
Stochastic gradient boosting	C/R; subsampled stagewise trees	sklearn available	L/T/W; subsampling does not make RF-GAP valid	Explicit boosted relation only	**reference-only**; historical bridge only.
XGBoost	C/R; scalable boosted partitions	Optional external Python package	L/T/W; Apache-2.0, actively maintained	Tree-contribution/leaf embedding, never ordinary RF-GAP	**specialty** initial boosted representative after relation semantics are defined.
LightGBM	C/R; leaf-wise histogram GBDT	Optional external Python package	L/T/W; MIT, maintained	Explicit boosted relation only	**specialty** for scale hypotheses, not a default comparison.
CatBoost	C/R; ordered categorical/oblivious trees	Optional external Python package	L/T/W; Apache-2.0, maintained	Explicit boosted relation only	**specialty** for mixed/categorical hypotheses.
Early oblique induction	C/R; learned hyperplane cuts	No maintained Python implementation found	L/T if supplied; historical implementation unavailable	Original co-leaf only	**reference-only**; OC1/modern oblique forests cover the comparison.
OC1	C/R; hill-climbed hyperplanes	Archived/third-party wrapper only	L/T; GPL wrapper, not a production dependency	Original co-leaf only	**reference-only** unless its particular optimization becomes a hypothesis.
Menze obliqueRF	C/R; regularized linear node splits	Archived R package	L/T; archived, avoid vendoring	Original co-leaf only	**reference-only**; retain as an R conformance/reference source.
Rotation Forest	C/R; PCA-rotated coordinate trees	aeon adapter available when installed	L/T; bootstrap/OOB only after adapter verification; aeon BSD-3-Clause, maintained	Original; RF-GAP only if IB/O contract is demonstrated	**reuse** and stabilize as projection-forest comparator.
Canonical Correlation Forest	C/R; supervised CCA hyperplanes	Native sklearn-compatible C/R estimators; deterministic fixture tied to the author MATLAB oracle	L/T established; CCF-Bag IB/O/W semantics remain gated	Original only; OOB/RF-GAP remain disabled	**modernized for original proximity.** Packaged C/R oracle checks pass; broader CCF-Bag weight semantics require a separate gate.
SPORF / treeple ObliqueRF	C/R; sparse random projection hyperplanes	Optional treeple adapter available	L/T/IB/O/W; treeple noncommercial, maintained research dependency	Original/OOB/RF-GAP if contract exposes bootstrap state	**reuse** and stabilize; optional/unavailable rows stay visible.
Isolation Forest	U; random axis-aligned isolation paths	sklearn `IsolationForest` available	Path scores native; L/T only for a separately defined collision relation; BSD-3-Clause	Native `score_samples`; any co-leaf relation must be labeled distinct	**reuse** native scores; **restricted** to native path-score baselines for anomaly claims.
Extended Isolation Forest	U; random oblique isolation paths	Optional treeple family / external EIF	Path scores native; L/T distinct from RF bootstrap	Native path score; explicit partition relation only	**reuse** native score when installed; no RF-GAP semantics.
Shi–Horvath unsupervised RF (paper)	U; observed-vs-synthetic classifier partitions	Native `SyntheticNullForest` meta-estimator	L/T/IB/O with persisted generator and seed	Original/OOB and native observed-class anomaly score	**integrated.** Relations index observed rows only; the synthetic

Foundational family	Task / partition geometry	Python/sklearn status; other-language oracle	Leaf/OOB contract; license & maintenance	Scientifically valid relation(s)	Decision and rationale
					construction is retained as provenance.
Quantile Regression Forest	R; conditional leaf distributions	Optional `quantile-forest` adapter	L/T and leaf-mass conditional weights; BSD-3-Clause, maintained	`conditional` weights plus native quantiles	**integrated.** Held-out tutorial reports pinball loss and interval coverage against native predictions.
Random Survival Forest	S; censored-outcome forest	Optional `scikit-survival` adapter; R `randomForestSRC` oracle	L/T plus native survival/risk/hazard outputs	Separately named `survival_leaf` relation	**integrated.** Evaluation uses training-derived IPCW concordance and integrated Brier score inside follow-up support.
Causal Tree	CA; honest treatment-effect partition	R `causalTree`; no current Python adapter	Honest structure/estimation masks essential	Honest neighborhood only with declared causal protocol	**specialty**; never infer causality from generic proximity.
Causal Forest	CA; honest adaptive treatment neighborhoods	R `grf` and Python causal comparators	Honest masks, treatment/outcome roles, W; GPL-3.0 R oracle	Matching/balancing and causal forest weights	**specialty** after leakage and estimand protocol.
Generalized Random Forest	CA/general local moments	R `grf` oracle	Same honest/role contract as causal forests	Local-moment forest weights	**specialty**; compare with Python causal tools, not generic RF-GAP.
Bayesian CART	C/R; posterior over partitions	No maintained exact adapter found	Posterior tree draws required	Posterior co-clustering only after semantics defined	**reference-only**.
BART / PyMC-BART	C/R; posterior additive trees	Optional PyMC-BART; R BART oracle	Posterior draws and tree contributions, not RF OOB	Posterior co-clustering/uncertainty relation	**specialty** only after posterior relation semantics and invariants are specified.

Relation constructors, independent of the estimator choice

Relation	Current status	Validity gate	Next action
Original co-leaf affinity	Available	L/T	Keep as symmetric baseline.
OOB co-leaf approximation	Available	Recoverable bootstrap O masks	Keep labeled as the current separable approximation, not a generic OOB formula.
RF-GAP	Available	Exchangeable bootstrap forest with L/IB/O/W; unsymmetrized for prediction identity	Keep as directed prediction weights; reject it for boosted/Bayesian/isolation ensembles.
Honest-GAP	Available for HonestForest classifier	Honest structure/estimation roles plus compatible OOB output	Re-run and report it; mark older ordinary-RF-GAP mismatch as legacy.
KeRF / leaf-mass normalization	Available as `kerf`	L/T and leaf mass	Symmetry/PSD and sparse/dense invariants are tested.
Path/branch similarity	Available as `path`	Decision paths and explicit normalization	Shared-path feature construction is tested as a PSD kernel.
Random-height partition kernel	Available as `random_height`	Decision paths plus a seeded cached cut protocol	Seed determinism, cache behavior, and PSD invariants are tested.
PF-GAP / structured-object relation	Missing	Object metric, split semantics and a structure-preserving evaluator	**Specialty** after vector-valued core benchmarks.

Application decision matrix

An application is a reusable use case, not one row per domain paper. Evidence links are representative primary sources; the literature ledger holds the full work-level evidence. “Current” records the actual package surface, runner, and test state at the freeze; a dash means not yet implemented, not “not useful.”

Tier / reusable use case	Representative evidence; required relation	Leakage/supervision risk and data	Baselines and metrics	Current API / runner
0 — prediction identity	RF-GAP; directed RF-GAP/Honest-GAP	Train/OOB/query roles must not mix; C/R labels	Native prediction/OOB output; max absolute error, row-weight mass	`prox_predict`, `prox_e` `run_prediction_iden` `test_prox.py`, `test_h`
0 — imputation	yaImpute; donor weights	Fit preprocessing/forest on training data only; MCAR/MAR/MNAR and feature types	Simple/iterative imputation; numeric RMSE, categorical accuracy, downstream C/R utility	`impute`; `run_imputati` `test_missingness.py`
0 — anomaly, novelty, mislabel audit	Isolation Forest, Shi–Horvath; native paths or declared relation	Do not use labels to score nominal unsupervised anomalies; contamination protocol	Native IF/EIF scores, supervised baselines; ROC-AUC, AP, rank/top-k agreement	`outlier_scores`; `run_outlier_detecti` tests are indirect
0 — local feature importance	Whitmore et al.; local relation perturbation	Evaluation labels cannot choose the explanation neighborhood	Permutation/conditional alternatives; local prediction drop, stability, runtime	`local_feature_impor` `run_local_feature_i` no dedicated unit suite
0 — prototypes and retrieval	Tree Space Prototypes; affinity or directed query weights	Prototype selection must not consume held-out labels; class constraints explicit	Random/class medoids/kNN; coverage, class purity, retrieval precision, stability	`get_prototypes`; `run_prototypes.py`; r
0 — manifold visualisation	RF-PHATE; affinity→distance or PHATE affinity	Labels only for colouring/evaluation; directed affinities need declared symmetrisation	PCA/UMAP/t-SNE/PHATE; trustworthiness, neighborhood preservation, label-free structure	`embed`, `embed_transfo` `run_mds_visualizati` `test_manifold.py`
1 — kernel SVM/KRR/kNN prediction	Scornet; symmetric PSD kernel or directed kNN weights	Split before kernel construction; train–test rectangular extension	Native forest, RBF/linear kernels, kNN; C/R score, calibration, PSD diagnostics	`ProximityKernelPred` `knn_predict` / method application tests
1 — clustering	Shi–Horvath; symmetric affinity/dissimilarity	Synthetic-null and cluster labels never share tuning roles	k-means/spectral/hierarchical; ARI/NMI/silhouette and stability	`proximity_clusterin` matrix runner / applica
1 — conditional distributions and quantiles	Meinshausen; leaf distribution weights	Conditional calibration evaluated on held-out data	Quantile forest/native quantile models; pinball loss, coverage, CRPS	`conditional_distrib` `conditional_quantil` adapter tests
1 — conformal/selective prediction	Yang et al. 2009; calibration scores plus declared neighborhood	Strict train/calibration/test separation; no label leakage through proximity tuning	Split conformal, RF intervals; marginal/conditional coverage, set size, risk–coverage	`split_conformal_cla` `split_conformal_reg` application tests
1 — batch active learning	Kee et al.; diversity/uncertainty relation	Pool labels unavailable until acquisition; batch redundancy control	Random/uncertainty/core-set; learning curve, area under curve, diversity	`select_active_batch` tests

Tier / reusable use case	Representative evidence; required relation	Leakage/supervision risk and data	Baselines and metrics	Current API / runner
1 — training-instance attribution	AXIL / RF-GAP; directed query-to-training weights	Attribution target and retraining counterfactual cannot be reused for selection	Influence/leave-one-out; deletion correlation, fidelity, sign/mass diagnostics	`instance_attributio` tests
1 — retrieval	Corporate-bond retrieval; query-to-index relation	Relevance judgments and labels held out by query/group/time	Cosine, Euclidean, kNN; Recall@k, mAP, nDCG, latency	`retrieve_neighbors`, `retrieval_metrics` / r runner / application tes
1 — proximity-neighborhood feature selection	Shi–Horvath; local stability/conditional relation	Feature selection nested within CV; no outcome leakage into neighbor graph	Filter/wrapper/embedded methods; selected-feature stability and C/R utility	`neighborhood_featur` `select_features` / ap
2 — personalized local models	Lee 2017; query weights	Patient/group/time leakage; local sample-size floor	Global RF, kNN, local GLM; per-case loss, calibration, subgroup error	— / — / —
2 — causal matching and balancing	Zhao et al.; honest/causal neighborhood	Treatment, outcome, and post-treatment variables strictly separated	Propensity/nearest-neighbor/causal forest; balance, ATT/ATE bias, coverage	— / — / —
2 — survival stratification	RSF clustering; survival relation	Censoring and time-dependent split leakage	RSF risk, Cox, clinical strata; C-index, IBS, time-dependent AUC, log-rank	— / — / —
2 — process/change monitoring	Auret–Aldrich 2011; temporal neighborhood/dissimilarity	Respect time order, delayed labels, and drift windows	Control charts, PCA, IF; detection delay, false-alarm rate, ARL	— / — / —
2 — multiview fusion/alignment	Cao et al.; per-view/cross-view relation	View pairing, batch and label leakage	Early/late fusion, CCA; task score, alignment, view ablation	— / — / —
2 — time series	Forest Proximities for Time Series; PF-GAP/structured relation	Temporal blocking and no future observations in features	DTW/distance forest, forecasting baselines; forecast error, anomaly delay	— / — / —
2 — arbitrary structured objects	generalized proximity forest; object-aware partition relation	Representation/metric is part of the estimator and evaluation	Object kernels/metric learners; task score, compute, invariance	— / — / —
3 — proximity graphs/GNNs	RF-GNN; sparse symmetric graph	Transductive edges and test labels/features must be controlled	kNN graph/GNN/MLP; C/R score, graph sparsity, inductive transfer	— / — / —
3 — autoencoding/denoising/compression	RF autoencoders; relation or path representation	Reconstruction target must not leak evaluation samples	PCA/AE/VAE; reconstruction, downstream utility, compression, robustness	— / — / —
3 — open domain case studies	Biology, medicine, finance, astronomy, remote sensing; use only a representative open set	Domain-specific group/time/censoring leakage	Domain-standard plus generic baselines; preregistered utility and failure modes	Existing runners use O
Deferred — label propagation	Evidence gap for direct forest-induced relation use	Graph labels can leak through edges	Graph SSL baselines; accuracy/calibration	— / — / —

Tier / reusable use case	Representative evidence; required relation	Leakage/supervision risk and data	Baselines and metrics	Current API / runner
Deferred — proximity resampling	Evidence gap	Synthetic labels/rare outcomes and resampling must be nested	SMOTE/class weighting; minority utility/calibration	— / — / —
Deferred — fairness-neighborhood audit	Evidence gap	Protected attributes, group membership and proxy leakage	Group calibration/error/balance; disparity confidence intervals	— / — / —

Repository and experimental sequence

Phase	Dependencies / blockers	Status	Measurable acceptance criterion
1. Evidence freeze	Targeted backward/forward searches from conformal, active-learning, monitoring, survival, causal, personalized, and selection seeds; URL/version/license/oracle checks	In progress: literature snapshot is strong	Two snowball passes add no reusable theme; every method/application/theme has exactly one decision and rationale.
2. Correctness and provenance	Shared `proximity_to_distance`; HonestForest optional dependency; legacy artifact location	Complete for 1.0	Sparse zeros map to maximal distance; honest identity uses `honest_gap`; pre-1.0 MDS artifacts are under `results/legacy`; the shared harness records provenance and failures.
3. Adapter architecture	Composition API and formal capability contract	Complete for registered 1.0 families	Adapters declare L/T/task/IB/O/W; invalid pairs fail explicitly; relation parity/invariant tests pass.
4. Core expansion	CCF oracle validation, relation specifications	Complete for the scoped 1.0 relations	Packaged C/R CCF fixture checks, KeRF/path/random-height invariants, and optional quantile/survival checks pass; CCF-Bag remains explicitly gated.
5. Tier 1 applications	Stable relations, deterministic shared splits, benchmark protocols	Utilities complete; broader benchmark studies remain future work	All listed utilities have focused tests; prediction/clustering/retrieval share the method-matrix runner and quantile has a held-out native comparison.
6. Specialty tracks	Data-specific protocol written and reviewed first	Deferred	Causal, survival, time-series, multiview, structured, graph, and autoencoding tracks individually satisfy their table acceptance criteria.
7. Cross-method study	Validity matrix and shared experiment harness	Harness and local matrix complete; full-scale estimation pending	The runner retains every valid/unsupported cell and records parent/downstream scores, diagnostics, revision/config/split, runtime, memory, and failures.

Required protocol for the cross-method study

Reuse the identical split and fitted parent model when relation constructors or applications are compared. Stratify datasets by axis-aligned versus rotated/oblique signal, `p/n`, feature correlation, mixed/categorical values, imbalance, missingness, cluster structure, censoring, temporal dependence, multiview structure, and non-vector objects. Do not pool raw scores across datasets: estimate within-dataset utility changes and method–data interactions.

Every run must record code revision, configuration, data/split identifiers, wall-clock runtime, peak memory, relation diagnostics, and failure/unsupported status. Future gates are adapter conformance, sparse/dense parity, relation-specific invariants, deterministic shared splits, absent-dependency behavior, CLI smoke tests, and application-specific leakage checks.

Immediate TODOs

1. Reconcile the literature ledger's work counts, bibliography keys, URLs, and lawful PDF inventory; retain “not found” as a bounded search result.
2. Create a capability registry that exposes L/T/task/IB/O/W and prevents invalid RF-GAP use on boosted, Bayesian, and isolation ensembles.
3. Replace `plot_mds`'s legacy sparse conversion with the tested shared conversion, archive current MDS conclusions as legacy, and rerun the Tier 0 smoke suites with provenance capture.
4. Re-run HonestForest only with `honest_gap` and label the older ordinary RF-GAP result as legacy.
5. Modernize CCF against its MATLAB oracle; then add KeRF, path/branch, and random-height constructors before broader application work.